

3.8 Maxima and Minima of Functions of Two Variables Worksheet

1. True or False: Every function has at least one local maximum.

False. Consider the function $f(x, y) = x + 1$. This function is a plane which has no local maximum or minimum.

2. Suppose that $z = f(x, y)$ is a differentiable function and $D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$. Determine if each of the following are True or False.

- If f has a local maximum or minimum at (a, b) , then it always satisfies $f_x(a, b) = f_y(a, b) = 0$.

True. All local maximums and minimums happen at critical points and if $f_x(a, b) = f_y(a, b) = 0$ then this means that (a, b) is a critical point. So, there must be a local maximum or minimum at (a, b) .

- If $f_x(a, b) = f_y(a, b) = 0$, then f has a local maximum or minimum at (a, b) .

False. All local maximums and minimums happen at critical points, however not every critical point corresponds to a local maximum or minimum. Since $f_x(a, b) = f_y(a, b) = 0$ we know that (a, b) is a critical point, but we have no guarantee that there is a minimum or maximum here.

For example, consider the function $f(x, y) = x^2 - y^2$ which has a saddle point at the critical point of $(0, 0)$.

- If f has a local minimum at (a, b) , then $f_{xx}(a, b) > 0$ and $D > 0$.

False. We know that if a local minimum occurs at (a, b) if $f_{xx}(a, b) > 0$ and $D > 0$, however it might also occur if the 2nd derivative test is inconclusive.

For example, $f(x, y) = x^4 - y^4$ has a local minimum at $(0, 0)$ but $f_{xx}(0, 0) = 0$ & $D = 0$.

3. Find all critical points of the function $f(x, y) = yxe^{-(x^2+y^2)}$.

We calculate that $f_x(x, y) = ye^{-(x^2+y^2)} + xye^{-(x^2+y^2)}(-2x) = ye^{-(x^2+y^2)}[1 - 2x^2]$ &

$$f_y(x, y) = xe^{-(x^2+y^2)} + xye^{-(x^2+y^2)}(-2y) = xe^{-(x^2+y^2)}[1 - 2y^2]$$

So, we need to solve the system $\begin{cases} ye^{-(x^2+y^2)}[1-2x^2]=0 \\ xe^{-(x^2+y^2)}[1-2y^2]=0 \end{cases}$.

The first equation will equal zero if $y=0$ or if $x=\pm\frac{1}{\sqrt{2}}$.

The second equation will equal zero if $x=0$ or if $y=\pm\frac{1}{\sqrt{2}}$.

Therefore, both equations will be satisfied at the points $(0,0)$ & $\left(\pm\frac{1}{\sqrt{2}},\pm\frac{1}{\sqrt{2}}\right)$. In total this means that f has 5 total critical points.

4. Find the critical points of $f(x,y)=5+6x-x^2+xy-y^2$ and classify them as local maxima, local minima, or saddles points.

We calculate that $f_x(x,y)=6-2x+y$ & $f_y(x,y)=x-2y$.

We need $\begin{cases} 6-2x+y=0 \\ x-2y=0 \end{cases} \rightarrow x=2y \rightarrow 6-2(2y)+y=0 \rightarrow y=2 \rightarrow x=4$. Therefore, we have a critical point at $(4,2)$.

We calculate $f_{xx}(x,y)=-2$ & $f_{yy}(x,y)=-2$ & $f_{xy}(x,y)=1$. Clearly these are all continuous functions.

We then calculate that $D(4,2)=(-2)(-2)-(1)^2=3$. Since $D>0$ & $f_{xx}(x,y)<0$ we know that the surface of $f(x,y)$ has a local maximum at the point $(4,2)$.

5. Let $f(x,y)$ be a function of two variables and suppose (a,b) is a critical point of f . Furthermore, suppose:

- $f_{xx}(a,b)=1$
- $f_{xy}(a,b)=2$
- $f_{yy}(a,b)=7$

Is $f(a,b)$ a local minimum, local maximum, or neither?

Clearly $f_{xx}(x,y)$ & $f_{yy}(x,y)$ & $f_{xy}(x,y)$ are all continuous functions.

We then calculate that $D(a,b)=(1)(7)-(2)^2=3$. Since $D>0$ & $f_{xx}(x,y)>0$ we know that the surface of $f(x,y)$ has a local minimum at the point (a,b) .

6. Find the local maximum and minimum values and saddle points of the function $f(x, y) = x^3 - 12xy + 8y^3$.

We calculate that $f_x(x, y) = 3x^2 - 12y$ & $f_y(x, y) = -12x + 24y^2$.

$$\text{We need } \begin{cases} 3x^2 - 12y = 0 \\ -12x + 24y^2 = 0 \end{cases} \rightarrow x = 2y^2 \rightarrow 3(2y^2)^2 - 12y = 0 \rightarrow 12y^4 - 12y = 0$$

$$\rightarrow 12y(y^3 - 1) = 0 \rightarrow y = 0 \text{ or } y = 1$$

We then see that if $y = 0$ then $x = 0$. Also, if $y = 1$ then $-12x + 24 = 0 \rightarrow x = 2$.

Therefore, we have a critical points of $(0, 0)$ & $(2, 1)$.

We calculate $f_{xx}(x, y) = 6x$ & $f_{yy}(x, y) = 48y$ & $f_{xy}(x, y) = -12$. Clearly these are all continuous functions.

We then calculate that:

$D(0, 0) = (0)(0) - (-12)^2 = -144$. Since $D < 0$ we know that the surface of $f(x, y)$ has a saddle point at $(0, 0)$.

$D(2, 1) = (12)(48) - (-12)^2 = 576 - 144 = 432$. Since $D > 0$ & $f_{xx}(x, y) > 0$ we know that the surface of $f(x, y)$ has a local minimum at the point $(2, 1)$.

7. Find the critical points of $f(x, y) = x^4 + 4xy + xy^2$ and classify them as local maxima, local minima, or saddles points.

We calculate that $f_x(x, y) = 4x^3 + 4y + y^2$ & $f_y(x, y) = 4x + 2xy$.

$$\text{We need } \begin{cases} 4x^3 + 4y + y^2 = 0 \\ 4x + 2xy = 0 \end{cases} \rightarrow 2x(2 + y) = 0 \rightarrow x = 0 \text{ or } y = -2.$$

We then see that if $x = 0$ then $4y + y^2 = 0 \rightarrow y(4 + y) = 0 \rightarrow y = 0 \text{ or } y = -4$.

So $(0, 0)$ and $(0, -4)$ are critical points.

If $y = -2$ then $4x^3 - 8 + 4 = 0 \rightarrow 4x^3 - 4 = 0 \rightarrow x = 1$.

So $(1, -2)$ is another critical point.

We calculate $f_{xx}(x, y) = 12x^2$ & $f_{yy}(x, y) = 2x$ & $f_{xy}(x, y) = 4 + 2y$. Clearly these are all continuous functions.

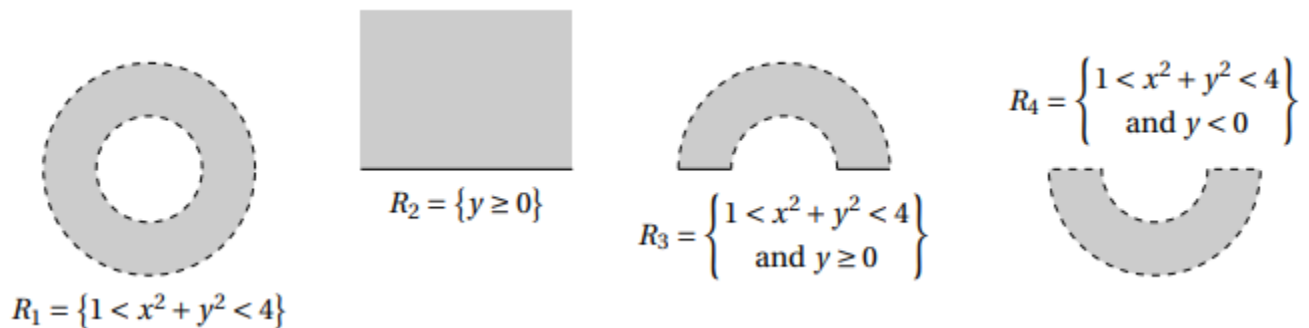
We then calculate that:

$D(0, 0) = (0)(0) - (4)^2 = -16$. Since $D < 0$ we know that the surface of $f(x, y)$ has a saddle point at $(0, 0)$.

$D(0, -4) = (0)(0) - (-4)^2 = -16$. Since $D < 0$ we know that the surface of $f(x, y)$ has a saddle point at $(0, -4)$.

$D(1, -2) = (12)(2) - (0)^2 = 24$. Since $D > 0$ & $f_{xx}(x, y) > 0$ we know that the surface of $f(x, y)$ has a local minimum at the point $(1, -2)$.

8. Consider the following four regions in the plane.



a) Classify each region as either open, closed, or neither.

R_1 is open since it doesn't include its boundary.

R_2 is closed since it does include its boundary.

R_3 is neither open nor closed since it only includes part of its boundary.

R_4 is open since it doesn't include its boundary.

b) Classify each region as either bounded or unbounded.

R_1 is bounded since it doesn't extend forever in any direction.

R_2 is unbounded since it extends infinitely in the horizontally and in the positive vertical direction.

R_3 is bounded since it doesn't extend forever in any direction.

R_4 is bounded since it doesn't extend forever in any direction.

9. Let $f(x, y)$ be a differentiable function on the disk $D = \{(x, y) | x^2 + y^2 = 100\}$ in $\mathbb{R}^2$, where

- $f(x, y) = 20$ for every point (x, y) on the circle $x^2 + y^2 = 100$.
- $f(0, 0) = 5$
- f has only one critical point, which is at $(1, 2)$.

Decide which of the four statements below is true.

A. $f(1, 2) > 5$

B. $f(1, 2) < 5$

C. $f(1, 2) = 5$

D. The relationship between $f(1, 2)$ & 5 cannot be determined from the given information.

The statement B is correct.

We know that since we are working on a closed and bounded region D the function f will obtain an absolute maximum and minimum value on D by the Extreme Value Theorem. By the first two bullet points we know that f does not achieve its absolute minimum on the boundary since for any (x, y) on the boundary of D we are told that $f(x, y) = 20$ and $f(0, 0) = 5$. This then means that f must obtain its absolute minimum at a critical point. Since the only critical point is at $(1, 2)$ the minimum must be obtained here which implies that $f(1, 2) < 5$.

10. Find the absolute maximum and minimum values of $f(x, y) = x^2 + 4y^2 - 2x^2y + 4$ on the rectangle given by $D = \{(x, y) \mid -1 \leq x \leq 1 \text{ \& } -1 \leq y \leq 1\}$.

First, we will draw a picture of the set D . We easily see that D is closed and bounded. So, by the EVT we know that an absolute min and max occur on D .

We calculate that $f_x(x, y) = 2x - 4xy$ and $f_y(x, y) = 8y - 2x^2$.

To find the critical points we need to solve the system:
$$\begin{cases} 2x - 4xy = 0 \\ 8y - 2x^2 = 0 \end{cases}$$

The first equation implies that $2x(1 - 2y) = 0 \rightarrow x = 0$ or $y = \frac{1}{2}$.

If $x = 0$ then by the second equation $y = 0$.

If $y = \frac{1}{2}$ then by the second equation $4 - 2x^2 = 0 \rightarrow x = \pm\sqrt{2}$.

Therefore, we have three critical points: $(0, 0)$, $(-\sqrt{2}, \frac{1}{2})$, & $(\sqrt{2}, \frac{1}{2})$.

However, $(0, 0)$ is the only critical point in D and is therefore the only critical point we must consider. We see that $f(0, 0) = 4$.

Now we must check for extremes along the boundary of D .

Along the segment C_1 the function f becomes $f(x, -1) = x^2 + 4 + 2x^2 + 4 = 3x^2 + 8$ where $-1 \leq x \leq 1$. We know that this is a parabola that opens upward with vertex occurring at $x = 0$. Therefore, the extreme values along C_1 are $f(0, -1) = 8$ and $f(-1, -1) = f(1, -1) = 11$.

Along the segment C_2 the function f becomes $f(-1, y) = 1 + 4y^2 - 2y + 4 = 4y^2 - 2y + 5$ where $-1 \leq y \leq 1$.

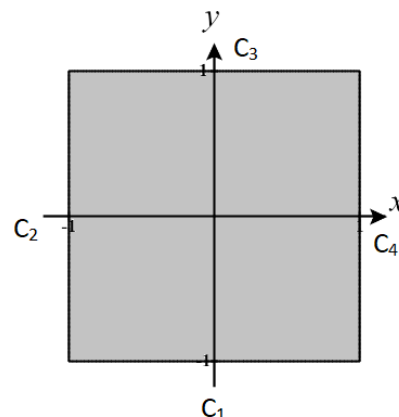
We know that this is a parabola that opens upward with vertex occurring at $y = \frac{1}{4}$. Therefore, the extreme values along C_2 are $f(-1, \frac{1}{4}) = 4.75$ and $f(-1, -1) = 11$.

Along the segment C_3 the function f becomes $f(x, 1) = x^2 + 4 - 2x^2 + 4 = 8 - x^2$ where $-1 \leq x \leq 1$. We know that this is a parabola that opens downward with vertex occurring at $x = 0$. Therefore, the extreme values along C_3 are $f(0, 1) = 8$ and $f(-1, 1) = f(1, 1) = 7$.

Along the segment C_4 the function f becomes $f(1, y) = 1 + 4y^2 - 2y + 4 = 4y^2 - 2y + 5$ where $-1 \leq y \leq 1$. We

We know that this is a parabola that opens upward with vertex occurring at $y = \frac{1}{4}$. Therefore, the extreme values along C_4 are $f(1, \frac{1}{4}) = 4.75$ and $f(1, -1) = 11$.

We see that the absolute maximum value of f on D is 11 (occurring at $(1, -1)$ and $(-1, -1)$) and the absolute minimum value of f on D is 4 (occurring at $(0, 0)$)



11. Find the absolute maximum and minimum values of $f(x, y) = 2x^2 - y^2 + 6y$ on region

$$D = \{(x, y) \mid x^2 + y^2 \leq 16\}.$$

First, we will draw a picture of the set D . We easily see that D is closed and bounded. So, by the EVT we know that an absolute min and max occur on D .

We calculate that $f_x(x, y) = 4x$ and $f_y(x, y) = -2y + 6$.

To find the critical points we need to solve the system:
$$\begin{cases} 4x = 0 \\ -2y + 6 = 0 \end{cases}.$$

The first equation implies that $x = 0$ and the second implies that $y = 3$.

Therefore, we have a single critical point of $(0, 3)$. We see that

$$f(0, 3) = 9.$$

Now we must check for extremes along the boundary of D .

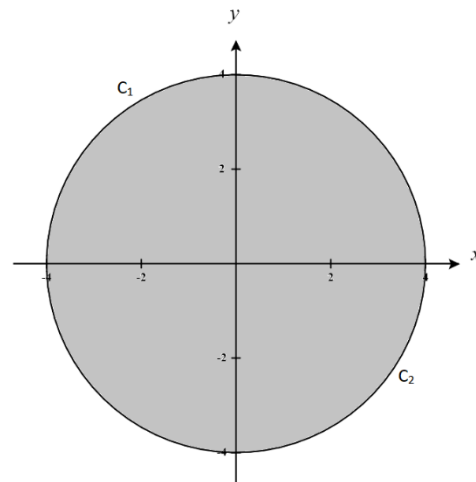
Along the segment C_1 (the left half of the circle) the function f becomes

$f(-\sqrt{16-y^2}, y) = 2(16-y^2) - y^2 + 6y = -3y^2 + 6y + 32$ where $-4 \leq y \leq 4$. We know that this is a parabola that opens downward with vertex occurring at $y = 1$. Therefore, the extreme values along C_1 are $f(-\sqrt{15}, 1) = 2(15) - 1 + 6 = 35$ and $f(0, -4) = -16 - 24 = -48$.

Along the segment C_2 (the right half of the circle) the function f becomes

$f(\sqrt{16-y^2}, y) = 2(16-y^2) - y^2 + 6y = -3y^2 + 6y + 32$ where $-4 \leq y \leq 4$. We know that this is a parabola that opens downward with vertex occurring at $y = 1$. Therefore, the extreme values along C_1 are $f(\sqrt{15}, 1) = 2(15) - 1 + 6 = 35$ and $f(0, -4) = -16 - 24 = -48$.

We see that the absolute maximum value of f on D is 35 (occurring at $(\pm\sqrt{15}, 1)$) and the absolute minimum value of f on D is -48 (occurring at $(0, -4)$)



12. Consider again the function $f(x, y) = 2x^2 - y^2 + 6y$ on region $D = \{(x, y) \mid x^2 + y^2 \leq 16\}$ as in #11.

a) Why would it be inconvenient to divide the boundary of D into two curves with formulas $y = \pm\sqrt{16-x^2}$?

If we use this division of the boundary then when we observe what the function becomes along each curve we will obtain $f(x, \pm\sqrt{16-x^2}) = 2x^2 - (16-x^2) + 6\sqrt{16-x^2}$.

While there is nothing *wrong* with this we can see that this function will *more difficult* to maximize and minimize.

- b) What would the function $f(x, y) = 2x^2 - y^2 + 6y$ need to look like so that dividing the boundary of D into two curves with formulas $y = \pm\sqrt{16-x^2}$ would be convenient to use?

There are many answers to this question. Below are a few options:

$$f(x, y) = 2x^2 - y^2 + 6x \quad (\text{This will make } f(x, \pm\sqrt{16-x^2}) = 2x^2 - (16-x^2) + 6x)$$

$$f(x, y) = 2x^2 - y^2 \quad (\text{This will make } f(x, \pm\sqrt{16-x^2}) = 2x^2 - (16-x^2))$$

Note that both of these options allow the resulting function to not contain any square roots.

13. Find the absolute maximum and minimum values of $f(x, y) = x^3 - 4xy^2$ on the triangle with vertices $(0, 0)$, $(4, 2)$, & $(-2, 2)$.

First, we will draw a picture of the set D . We easily see that D is closed and bounded. So, by the EVT we know that an absolute min and max occur on D .

We calculate that $f_x(x, y) = 3x^2 - 4y^2$ and $f_y(x, y) = -8xy$.

To find the critical points we need to solve the system:

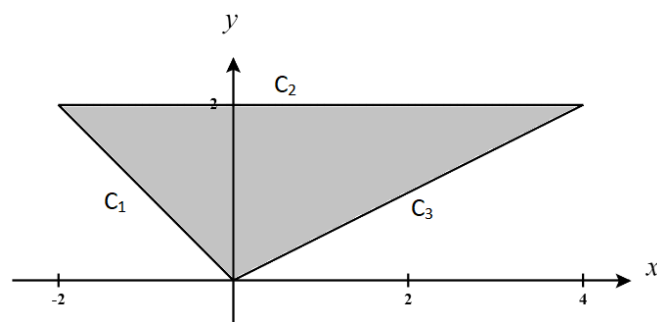
$$\begin{cases} 3x^2 - 4y^2 = 0 \\ -8xy = 0 \end{cases}$$

The second equation implies that $x = 0$ or $y = 0$.

If $x = 0$ then by the first equation $y = 0$. If $y = 0$ then by the first equation $x = 0$.

Therefore, we have only one critical point of $(0, 0)$. We see that $f(0, 0) = 0$

Now we must check for extremes along the boundary of D .



Along the segment C_1 (given by the equation $y = -x$) the function f becomes $f(x, -x) = x^3 - 4x^3 = -3x^3$ where $-2 \leq x \leq 0$. We can easily see that the extreme values along C_1 are $f(-2, 2) = 24$ and $f(0, 0) = 0$.

Along the segment C_2 (given by the equation $y = 2$) the function f becomes $f(x, 2) = x^3 - 16x$ where $-2 \leq x \leq 4$. Finding the extremes along C_2 will be more difficult to do than in previous examples since we have a more complicated function $f(x, 2) = x^3 - 16x$. We will use our knowledge from Calculus I to find absolute minimums and maximums of this curve. We see that:

$$f'(x, 2) = 3x^2 - 16$$

This function has critical values of $x = \pm\sqrt{\frac{16}{3}} \approx \pm 2.3$. Therefore, $x = \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}}$ is the only critical value in $-2 \leq x \leq 4$. When checking the outputs when compared with the endpoints we have:

$$f\left(\frac{4}{\sqrt{3}}, 2\right) = \frac{64}{3\sqrt{3}} - \frac{64}{\sqrt{3}} = \frac{-128}{3\sqrt{3}} \approx -24.6, \quad f(-2, 2) = 24, \quad f(4, 2) = 0.$$

So along C_2 the extremes are $f\left(\frac{4}{\sqrt{3}}, 2\right) = \frac{-128}{3\sqrt{3}}$ & $f(-2, 2) = 24$

Along the segment C_3 (given by the equation $y = \frac{1}{2}x$ the function f becomes $f\left(x, \frac{1}{2}x\right) = x^3 - x^3 = 0$ where $0 \leq x \leq 4$. Very clearly along C_3 the function f is always 0. Therefore, the extreme value along C_3 is $f\left(x, \frac{1}{2}x\right) = 0$ for all $0 \leq x \leq 4$.

We see that the absolute maximum value of f on D is 24 (occurring at $(-2, 2)$) and the absolute minimum value of f on D is $\frac{-128}{3\sqrt{3}}$ (occurring at $\left(\frac{4}{\sqrt{3}}, 2\right)$).